① srcML of lines 14 and 18–19

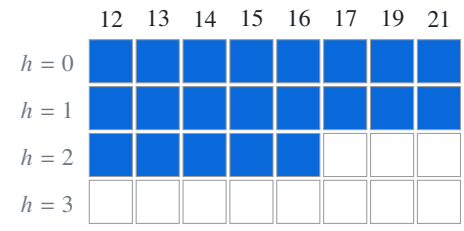
```
<expr_stmt pos:start="14:5">[g]<expr><call><name><name>qc</name><operator>.</operator><name>cx[c]</name></name>
    ↪ <argument_list>(<argument><expr><name><name>q[d]</name><index>[<expr><name>src</name></expr>]</index></name>
    ↪ </expr></argument>, <argument><expr><name><name>q</name><index>[<expr><name>a</name></expr>]</index></name>
    ↪ </expr></argument>)</argument_list></call></expr></expr_stmt>

<with pos:start="18:5">[g]with <expr><call><name><name>qc</name><operator>.</operator><name>if_test[f]</name>
    ↪ </name><argument_list>(<argument><expr><tuple>(<expr><name><name>c</name><index>[<expr>
    ↪ <literal type="number">2[e]</literal><operator>*</operator><name>h</name><operator>+</operator>
    ↪ <literal type="number">1</literal></expr>]</index></name></expr>, <expr><literal type="number">1</literal>
    ↪ </expr>)</tuple></expr></argument>)</argument_list></call></expr><block>:<block_content>
        <expr_stmt><expr><call><name><name>qc</name><operator>.</operator><name>x[c]</name></name>
            ↪ <argument_list>(<argument><expr><name><name>q[d]</name><index>[<expr><name>b</name></expr>]</index>
            ↪ </name></expr></argument>)</argument_list></call></expr></expr_stmt>
</block_content></block></with>
```

c gate **d** qubit argument, evaluated per iteration **e** classical bit, written by a measurement and read by the condition
f `if_test` → conditioned block **g** position → source line

② Trace, line 8 and hop $h = 0$ ⟶ ③ Slice of `q[4]`

line	hop	operation
8	–	u q_0
12	0	h q_1
13	0	cx q_1, q_2
14	0	cx q_0, q_1
15	0	h q_0
16	0	measure $q_0 \rightarrow c_0$
17	0	measure $q_1 \rightarrow c_1$
19	0	x q_2 if c_1
21	0	z q_2 if c_0
	1–3	the same on the next qubits



line 8 **in**, line 23 out **22** of 34 operations